

NCIE-009 Skeletal v0.1 → Full v1.0

Expansion Verification Report (Corrected)

Security, IAM & Zero-Trust Architecture — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

0. Pre-Approval Surgical Correction Record

This report supersedes the prior Verification Report, which contained a false statement: it certified that a programmatic scan confirmed an explicit '<N>.0 Salient Things to Note' heading in all 28 chapters, when Chapter 27 (Human Review Register & Open Security Decisions) in fact transitioned directly from its Status/Upstream blocks to '27.1 Purpose' with no '27.0 Salient Things to Note' subsection — a violation of the production directive's item 5 hard gate. This was identified during independent human examination (David King Boison, PhD — Academic Intelligence Center) of the delivered v1.0 package, not by this build's own QA pass, which is itself noted as a process gap corrected going forward.

CORRECTION APPLIED: A '27.0 Salient Things to Note' subsection was inserted immediately before '27.1 Purpose', using the already-approved skeletal Chapter 27 salient content verbatim ('Do not invent answers; Blocking scope must be precise and aligned with NCIE-008 authority; deduplicate identical upstream governance questions.') — no new security doctrine was introduced. Sections 27.1 through 27.6, all 28 Human Review items, and every other chapter are unchanged. This report's own claim in §3 was corrected to describe an actually-rerun scan rather than restate the prior (false) one. The specification version is unchanged at v1.0 — this is a pre-approval correction, not a v1.0→v1.1 amendment.

1. Scope Verification

NCIE-009 v1.0 is a controlled full expansion of the approved NCIE-009 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal security component, upstream dependency/authority and Human Review focus is elaborated into security doctrine, control matrices, RACI/control-ownership structure and Human Review items, without inventing IdP/PAM/KMS/HSM/SIEM/SOAR/DLP/network products, cloud services, cryptographic algorithms, or any session/token/assurance/rotation/patch/incident/retention threshold absent upstream authority.

2. 28-Chapter Integrity Verification

A programmatic scan of the final corrected document confirms exactly 28 sequential 'NCIE-009 — Chapter N Expansion' labels, N = 1..28, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

3. Skeleton Fidelity and Salient Things to Note Hard-Gate Verification (Rerun)

Each of the 28 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Security Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). Per directive item 5's hard gate, every chapter must retain an explicit Salient Things to Note subsection, rendered as an '<N>.0 Salient Things to Note' heading immediately after the chapter's Status/Upstream blocks. A programmatic scan of the corrected final document — matching the regular expression '^(\d+)\.0 Salient Things to Note\$' against every paragraph — was rerun against the actual file rather than assumed from the build script, and found this heading present for all 28 chapters (N = 1 through 28), with zero missing. Chapter 27's instance is the one added by this correction; the remaining 27 were already present and unchanged. No approved chapter concept was omitted, and none was conceptually inverted.

4. No Product, Algorithm or Threshold Invention

A full-text scan for named identity providers, PAM/KMS/HSM/SIEM/SOAR/DLP products, cloud services, or specific cryptographic algorithms found none invented (directive item 8) — Chapter 11 and Chapter 12 explicitly defer product and algorithm selection. A separate scan for session/token durations, assurance levels, factor sets, rotation periods, patch SLAs, incident severities, network zone counts, DLP thresholds, retention periods or test frequencies found none invented (directive item 9) — every such matter is instead recorded as a named Human Review item (e.g., HR9-5-1 for assurance/step-up, HR9-10-1 for session/credential classes, HR9-24-1 for remediation authority).

5. Governing Security Invariants Verification

The 16 invariants proposed in the approved Skeletal v0.1 ('Proposed Security Invariants') are restated unchanged in front matter §0.3, and are additionally cross-referenced — not restated — against the 12 NCIE-008 governance invariants and other upstream invariants already binding from NCIE-005/006/007. A full-text scan confirms every invariant is both stated once in §0.3 and applied by name in at least one operative chapter (e.g., AUTHENTICATED IDENTITY ≠ AUTHORIZED ACTION in Ch.5; ENCRYPTION ≠ AUTHORIZATION in Ch.12; BREAK-GLASS ≠ PERMANENT PRIVILEGE in Ch.9; RECOVERY ≠ REAUTHORIZATION in Ch.25).

6. Security/Governance Boundary Verification

Chapter 1 Table 1 fixes what NCIE-009 does not own (product requirements, architecture, canonical data, engineering baseline, ARGUS behavior, Memory/Context architecture, orchestration, and institutional governance/RACI/risk acceptance — all NCIE-001–008's). Every consequential security control in Chapters 5–26 is expressed as enforcement of a named NCIE-008 Chapter 4 decision-right class with the accountable holder marked 'to be assigned,' never as NCIE-009 assigning or reassigning RACI, approval ownership or risk-acceptance authority itself (directive item 7).

7. Identity Taxonomy and Non-Equivalence Verification

Chapter 3 Table 4 fixes seven principal classes (Human, workload, Agent, service, session/device, privileged, validator) each with its own authoritative source and an explicit 'never substitutes for' column. AGENT IDENTITY ≠ HUMAN IDENTITY and SERVICE IDENTITY ≠ HUMAN IDENTITY are restated in Chapters 3, 7, 10 and 14 wherever an Agent or service identity could otherwise be mistaken for a Human's.

8. Authentication and Authorization Verification

Chapter 5 confirms AUTHENTICATED IDENTITY ≠ AUTHORIZED ACTION and Table 7 fixes 5 assurance contexts without inventing specific factor sets. Chapter 6 confirms REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION and Table 9 fixes deny-by-default as the enforcement default across all four authorization layers (object/field/action/purpose).

9. Delegation, Privileged Access and Break-Glass Verification

Chapter 7 confirms DELEGATED TASK ≠ HUMAN CREDENTIAL DELEGATION and Table 11 binds every delegation class to an explicit scope, expiry and revocation trigger. Chapters 8–9 confirm SECURITY ADMINISTRATION ≠ SUBSTANTIVE CONTENT AUTHORITY and BREAK-GLASS ≠ PERMANENT PRIVILEGE respectively; Tables 13 and 15 (the two-part Privileged/Break-Glass Matrix) bind every privilege and emergency class to a named scope, an NCIE-008 approval interface or an independent post-event reviewer.

10. Session, Secrets and Cryptography Verification

Chapter 10 confirms human credentials are never copied into Agent Context and treats revocation as an immediate, cross-system event. Chapter 11 prohibits secrets in prompts, logs, Agent Context or source code as an absolute rule routed to Chapter 22 on violation, and separates key custody from ordinary application identity without naming a specific KMS/HSM product. Chapter 12 confirms ENCRYPTION ≠ AUTHORIZATION and defers specific algorithm selection to engineering/institutional decision (HR9-12-1, Non-Blocking).

11. Network, Agent Runtime and Model/Tool Security Verification

Chapter 13 confirms NETWORK REACHABILITY ≠ APPLICATION AUTHORIZATION with deny-by-default network policy. Chapter 14 confirms Agent Runtime Identity ≠ Human Identity and that Dynamic Agent Synthesis cannot expand permission, routing generated code through Chapter 23's security gate. Chapter 15 confirms MODEL ELIGIBILITY ≠ DATA DISCLOSURE AUTHORITY. Chapter 16 confirms TOOL ELIGIBILITY ≠ TOOL EXECUTION AUTHORITY and defines OUTCOME UNKNOWN reconciliation for ambiguous external-call outcomes.

12. Context, Data Classification and Privacy Security Verification

Chapter 17 confirms CONTEXT PROPAGATION ≠ AUTHORIZATION PROPAGATION and that SEMANTIC RETRIEVAL ≠ AUTHORITY TO DISCLOSE, requiring cached-Context reconciliation against current authorization. Chapter 18 confirms VPF PASS ≠ SECURITY AUTHORIZATION and Table 25's Data-Flow Security Matrix fixes a classification check point on all five named data paths (Tool, export, model input, voice output, Agent-to-Agent handoff). Chapter 19 requires inferred sensitive data to be classified and audited at the same standard as directly sourced protected data.

13. Logging, Detection and Incident Response Verification

Chapter 20 confirms LOGGING ≠ AUTHORITY TO DISCLOSE and preserves telemetry/audit/provenance as three distinct records. Chapter 21 confirms security analytics alerts do not automatically become institutional Findings (MODEL/AGENT OUTPUT ≠ EVIDENCE, FINDING OR DECISION). Chapter 22 confirms containment operates only under NCIE-008 Ch.22 governed emergency authority and that evidence/historical state are preserved through containment, with RECOVERY ≠ REAUTHORIZATION carried into the recovery handoff.

14. Supply-Chain, Vulnerability and Resilience Verification

Chapter 23 confirms verified artifact integrity does not itself create authorization and gates generated code separately from ordinary dependency scanning. Chapter 24 confirms severity classification alone does not create risk acceptance and routes every security exception through the NCIE-008 Ch.21 exception lifecycle (EXCEPTION APPROVAL ≠ POLICY REPEAL). Chapter 25 confirms RECOVERY ≠ REAUTHORIZATION at the resilience layer and treats an unverified backup/failover target as compromised until proven otherwise.

15. Security Testing and Test-Status Integrity Verification

Chapter 26 Table 34 specifies exactly the 9 test classes the directive names (IAM bypass, privilege escalation, cross-context leakage, sandbox escape, prompt injection, DLP bypass, Tool misuse, recovery/revocation failure, supply-chain compromise), each traced to the chapter(s) it verifies. Every test is marked TEST SPECIFIED only; a full-text scan confirms no test anywhere in this document is marked TEST EXECUTED, TEST PASSED, or claims IMPLEMENTATION EVIDENCE AVAILABLE (directive items 52 and 57) — detailed test-case authoring and execution are explicitly deferred to NCIE-016.

16. Threat-Control Traceability Verification

Chapter 2 Table 3 maps 7 material threats/abuse cases (credential theft, privilege escalation, sandbox escape, prompt injection, data exfiltration, supply-chain compromise, compromised-backup reactivation) to preventive, detective, responsive and recovery controls, each citing the chapter that owns the control (directive item 49). No threat in Table 3 is left with an empty control column.

17. Human Review Register Mechanical Reconciliation (Correct From First Build)

Chapter 27's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-26, plus HR9-27-1 (Chapter 27's own item) and HR9-28-1 (Chapter 28's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build (where an equivalent self-referential item was appended only as an afterthought and omitted from its own table). Result: 28 total items, 24 Blocking / 4 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 24 Institutional Confirmation Required, 3 Security/Sovereignty Confirmation Required (Chapters 14, 15, 19), 1 Resolved by Upstream Baseline (Chapter 28).

18. Upstream Human Review Deduplication (Directive Item 51)

HR9-1-1 (confirm NCIE-009 acceptance and chapter sign-off authority) restates the identical institutional question NCIE-008 already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Chapter 27 Table 37 cross-references HR8-10-1 rather than treating HR9-1-1 as an independent open question. Every other NCIE-009 Human Review item was individually checked against the NCIE-008 Human Review Register and found to concern a security-specific decision (an assurance level, a threshold, a control-ownership assignment, or a test/acceptance authority) that NCIE-008 does not itself resolve — no other duplicate was found.

19. Blocking Scope Discipline Verification

Each Blocking item in the Chapter 27 register states precisely which control default, threshold, ownership assignment or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 12, 20, 21, and the resolved Chapter 28 item) are explicitly marked as not gating the corresponding chapter's security operation in the interim.

20. Traceability and Security-Control-Ownership Verification

Chapter 28 Table 38 traces all 28 chapters to their primary upstream authority (approved NCIE-001-008 and/or the approved NCIE-009 Skeletal v0.1); no chapter states a security requirement without an identifiable source. Table 39 (Security-Control-Ownership Matrix) additionally maps 19 control domains to their owning chapter(s) and the NCIE-008 decision-right class governing institutional approval, so no control domain in Chapters 3-26 is left without a traceable governance hook. No approved NCIE-001-008 requirement, architecture element, canonical object, engineering baseline element, ARGUS behavior, Memory/Context architecture element, orchestration element or governance decision is redefined, reinterpreted or contradicted anywhere in this document (directive item 18, upstream non-redefinition).

21. Downstream Non-Preemption Verification

Chapter 28 Table 40 confirms this document hands over authentication/authorization and Tool-security contracts to NCIE-011 (APIs), step-up/break-glass UX requirements to NCIE-012, logical secrets/crypto/log-retention requirements to NCIE-014, security operational responsibilities to NCIE-015, and the 9 security test classes to NCIE-016 — without specifying their technical implementation, consistent with directive item 19 (downstream non-preemption).

22. No Fabricated Compliance or Implementation Claims

A full-text scan for legal, regulatory, ISO, NIST or other certification/compliance claims found none invented (directive item 56). A separate scan confirms no control anywhere in this document is claimed deployed, operational, penetration-tested, certified or production-approved merely because this v1.0 specifies it (directive item 57) — every control-ownership table marks its holder 'to be assigned,' and Chapter 26 explicitly marks every test SPECIFIED, not executed.

23. Version, Document-Control and Running-Header Verification

Cover and Document Governance table confirmed reading: Edition 'In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)'; Requirements/Architecture/Canonical-data/Engineering/ARGUS-behavioral/Memory-

Context/Orchestration/Governance authority rows all read their approved current versions (NCIE-001 v1.3 through NCIE-008 v1.0); a 'Prior baseline' row records the approved Skeletal v0.1 as historically preserved; Status reads IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The running header — rendered from a copied NCIE-008 v1.0 template to inherit this series' visual styling — was corrected to 'NCIE-009 | SECURITY, IAM & ZERO-TRUST ARCHITECTURE | VERSION 1.0' and the footer corrected from 'CONTROLLED GOVERNANCE DOCUMENT' to 'CONTROLLED SECURITY DOCUMENT' and from 'NCIE-008' to 'NCIE-009' before this document was rendered to PDF.

24. Full-Document Cross-Reference Scan

Every occurrence of 'NCIE-008' in the final document was individually inspected (147 occurrences outside Upstream/Trace blocks — far more than the small number of legitimate NCIE-007 citations found in the companion NCIE-008 document, because NCIE-009 sits immediately beneath NCIE-008 and enforces its decision rights pervasively). Each occurrence was confirmed to be a legitimate governance citation (e.g., 'NCIE-008 Ch.4 decision-right class', 'Accountable Authority Class (NCIE-008, to be assigned)') consistent with directive item 7's requirement that NCIE-009 consume, not redefine, NCIE-008 governance — none is a leftover donor-template branding error. The cover, header, footer and Document Governance 'Document' row were separately confirmed to read NCIE-009, not NCIE-008.

25. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python exec() call with an explicit encoding='utf-8' argument, a fix first required during the NCIE-008 build), a full-document scan of the corrected NCIE-009 document confirmed zero mojibake occurrences across all 587 paragraphs (585 + 2 for the corrected §27.0 subsection heading and body) and 41 tables (unchanged — no table was added or removed by this correction).

26. Visual and Structural Verification

The corrected document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with the new '27.0 Salient Things to Note' entry alongside the existing chapter/section titles and real page numbers, repaginated to 54 pages (unchanged from the prior build), saved, and rendered to PDF. The new §27.0 subsection was confirmed to render cleanly immediately after Chapter 27's Upstream block and before '27.1 Purpose,' with no clipping, overlapping text, or broken table anywhere in the security matrices, the Chapter 27 Human Review Register tables, or the Chapter 28 traceability/control-ownership/handover tables.

27. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR9-<chapter>-<n>: TOTAL IDS FOUND = 28. UNIQUE IDS = 28. DUPLICATES = 0. OPEN = 27. RESOLVED = 1 (HR9-28-1, Resolved by Upstream Baseline). BLOCKING = 24. NON-BLOCKING = 4. CLASSIFICATION TOTALS: Institutional Confirmation Required = 24; Security/Sovereignty Confirmation Required = 3; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 28 unique IDs found anywhere in the document also appears as a row in the Chapter 27 Complete Register table, which itself contains exactly 28 rows with no duplicate Review ID.

28. Three-Artifact Consistency Check (Corrected)

The corrected NCIE-009 v1.0 main document, this corrected Verification Report and the corrected Full-Expansion Change Register agree exactly on: Source Version v0.1 (Skeletal) → Target Version v1.0 (Full, unchanged by this correction — no version increment); Authority chain approved NCIE-001 v1.3 through approved NCIE-008 v1.0, plus the approved NCIE-009 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 28 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 28/28 explicit subsections present (corrected from 27/28); Human Review Register TOTAL = 28, BLOCKING = 24, NON-BLOCKING = 4, classification totals as in §27 above (unchanged by this correction — no Human Review item was added, resolved, reclassified or merged); orphan Human Review IDs = 0; duplicate Human Review IDs = 0; security test-class status = 9 tests specified, none executed or passed; generated-code status = not activated by NCIE-009, additional security/engineering confirmation still required (Ch.14, Ch.23); security invariants = 16 NCIE-009-specific plus cross-referenced upstream invariants. No mismatch was found between the three corrected artifacts.

29. Final Readiness Disposition

NCIE-009 — Security, IAM & Zero-Trust Architecture v1.0 — Full Expansion of Approved Skeletal v0.1 (Pre-Approval Corrected) — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed control default, threshold, ownership assignment and product/algorithm choice deferred in Chapters 1-26 remains a proposal pending resolution of the Chapter 27 Human Review Register; every §26.1 security test remains SPECIFIED, not executed, pending NCIE-016. The Salient Things to Note hard gate now passes 28/28 with no exception. This document is NOT marked Approved, and this correction did not increment its version.